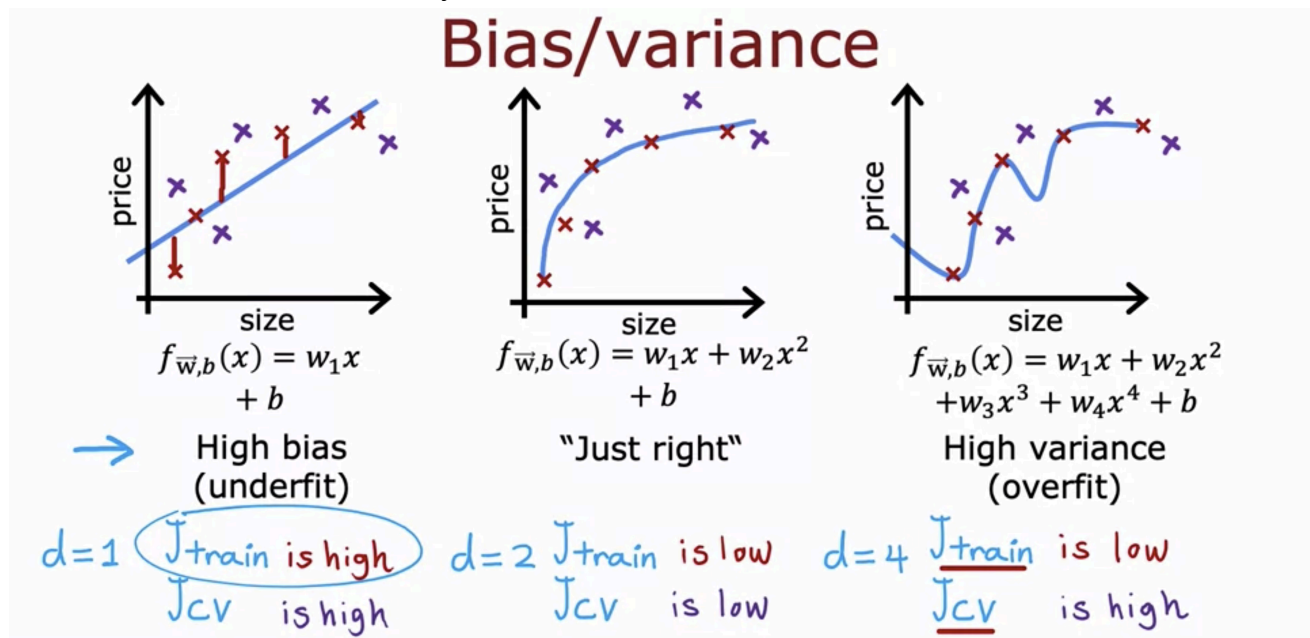


Iterative Loop of ML Development

- Choose architecture(model, data, etc)
- Train Model
- Diagnostics (Bias, Variance, Error Analysis)

Error Analysis

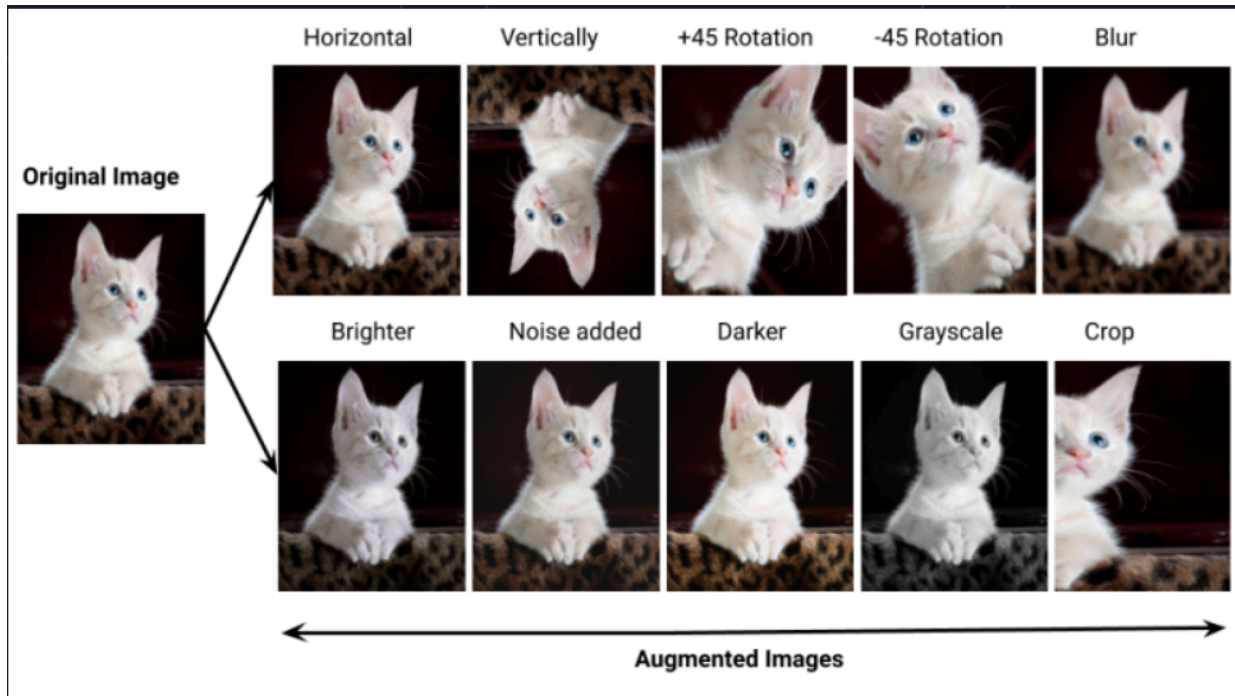
- Do the Bias and Variance Analysis



Adding Data

- Add more data where the error analysis is indicated
 - which is the larger problem
 - where the model is doing its worse
- Data Augmentation

- Modifying existing training examples to create more and new training examples



- Data Synthesis
 - Creating our own new training examples
- AI = Code + Data
 - Model Centered Approach : Focusing on code more
 - Data Centered Approach : Focusing on data more

Transfer Learning

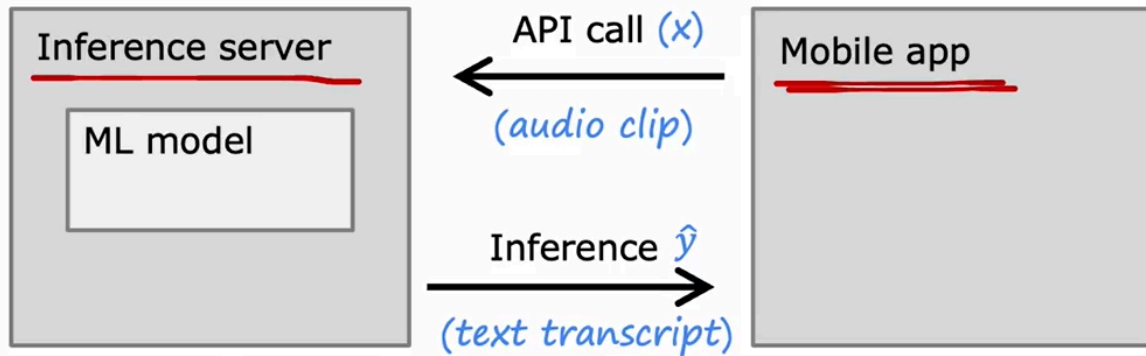
- machine learning technique where you **reuse a model trained on one task as the starting point for a different but related task.**
- Used when you have less training examples to train a model.
- Steps :
 - Supervised Pre-training (Training the first model)
 - You can you model from somewhere else but the input type should be same(images, audio, text)
 - Fine tuning (Training the other model)
- Options :
 - Option 1 : Train only the last layer
 - Option 2 : Train all the parameters initialized with the other model

Full Cycle of Machine Learning Project

- Define Project
- Collect Data

- Train Model
- Deploy in Production

Deployment



- MLOps
 - Practice of how to build and deploy a machine learning system.

Fairness, Bias, and Ethics

- Biased bank loan approvals
- Deepfake Videos
- Brainstorm things that might go wrong before starting a project
- Audit the system against possible harm prior to deployment
- Develop a mitigation plan

Error metrics for skewed datasets

- The error analysis doesn't work on these kinda datasets where the odds are far away from 50-50. (Ex : Rare Diseases)

Precision/recall

$y = 1$ in presence of rare class we want to detect.

		Actual Class	
		1	0
Predicted Class	1	True positive 15	False positive 5
	0	False negative 10	True negative 70
		↓ 25	↓ 75

Precision:

(of all patients where we predicted $y = 1$, what fraction actually have the rare disease?)

$$\frac{\text{True positives}}{\text{\#predicted positive}} = \frac{\text{True positives}}{\text{True pos} + \text{False pos}} = \frac{15}{15 + 5} = 0.75$$

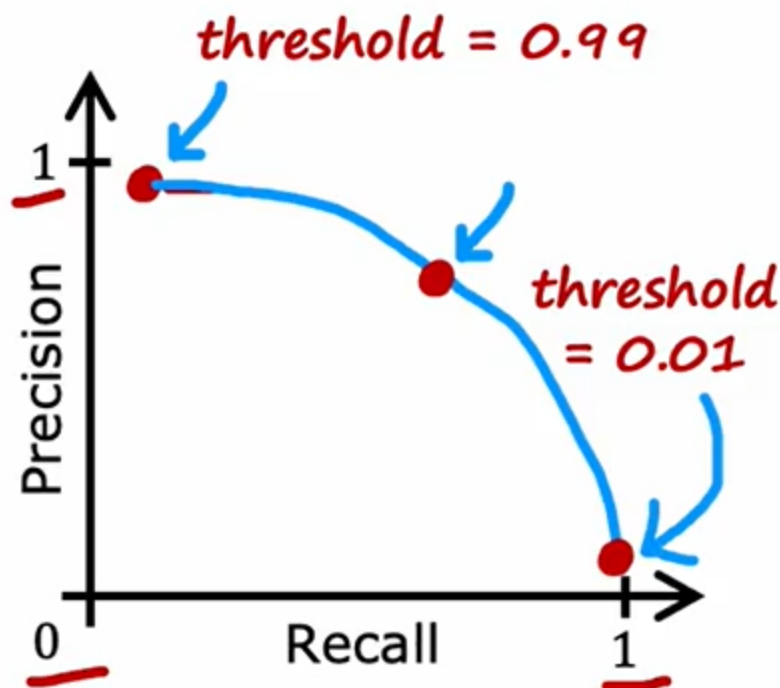
Recall:

(of all patients that actually have the rare disease, what fraction did we correctly detect as having it?)

$$\frac{\text{True positives}}{\text{\#actual positive}} = \frac{\text{True positives}}{\text{True pos} + \text{False neg}}$$

Trading off Precision and Recall

- If y thrush-hold is increased it increase precision and decreases recall and wise-versa.



- F1 Score
 - $2 (\text{Precision} \text{ Recall}) / (\text{Precision} + \text{Recall})$